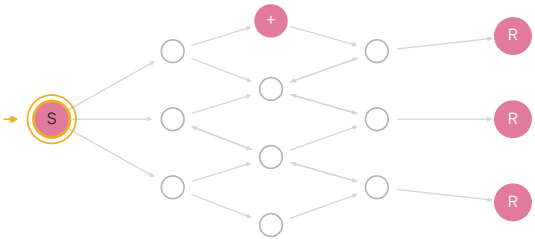
# Predictive risk and recording decisions in partially observed neural networks

Five calibration recordings | Held-out stimulus response | Known connectivity, uncertain gains

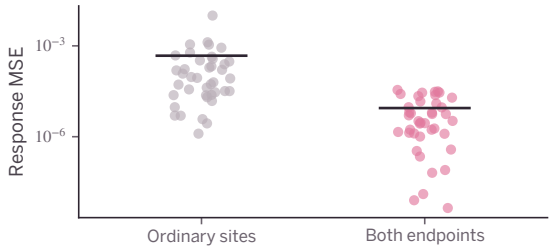
## a Record the stimulus and readouts




1 stimulus + 3 readouts + 1 additional site

100-unit network (schematic)

## b Improve response prediction




Endpoint observations improve held-out prediction.

40 independent networks; bars show means.

## c Locate the information mechanism

Input → Recurrent → Readout

Exact finite-response factorization

Endpoint observations constrain stimulus entry and readout, narrowing compatible responses.

Recurrent gain combinations retain additional information.

## d Use the remaining recording

Retain endpoints; target recurrent combinations.

Width² along one internal direction

Targeted / random fifth site; adjusted interval

**Record endpoints for prediction; target the remaining internal information.**